

$$E(X) = \sum_{i \in R(X)} iP(X = i) = \mu$$

$$Var(X) = E(X^2) - \mu^2 = \sigma^2$$

$$E(X)^2 = \sum_{i \in R(X)} i^2 P(X = i)$$

$$\text{B:} \begin{cases} R(x) &= \{0, 1, \dots, n\} \\ P(X = j) &= \binom{n}{j} p^j (1-p)^{n-j} \\ &: \text{Prob of } j \text{ succ. out of } n \text{ trials} \\ E(X) &= np = \mu \\ Var(X) &= \sqrt{np(1-p)} = \sigma \end{cases}$$

$$\text{Poi:} \begin{cases} R(x) &= \{0, 1, 2, 3, 4, \dots\} \\ P(X = j) &= \frac{e^{-\lambda} \lambda^j}{j!} \\ &: \text{Prob of } j \text{ occ. in interv.} \\ E(X) &= \lambda \\ Var(X) &= \lambda \quad (SD = \sqrt{\lambda}) \end{cases}$$

$$\text{G:} \begin{cases} R(x) &= \{1, 2, 3, 4, \dots\} \\ P(X = j) &= (1-p)^{j-1} p, \quad 0 < p < 1 \\ E(X) &= 1/p \\ Var(X) &= (1-p)/p^2 \end{cases}$$

$$\text{HG:} \begin{cases} R(x) &= \max(0, n+m-N), \dots, \min(m, n) \\ P(X = j) &= \frac{\binom{m}{j} \binom{N-m}{n-j}}{\binom{N}{n}} \\ E(X) &= \frac{nm}{N} \\ Var(X) &= \frac{nm(N-n)(N-m)}{N^2(N-1)} \end{cases}$$

$$\text{Co:} \begin{cases} f(x) &\geq 0 \forall x \in R(X) \\ \int_{R(X)}^a f(x) dx &= 1 \\ P\{a < X < b\} &= \int_a^b f(x) dx \\ E(X) &= \int_{R(X)} x f(x) dx = \mu \\ Var(X) &= E(X^2) - \mu^2 \\ f(X) &= a \neq \text{prob!} \end{cases}$$

$$\text{Exp:} \begin{cases} f(x) &= \lambda e^{-\lambda x} \text{ where } x \geq 0 \\ E(x) &= \frac{1}{x} \\ Var(x) &= \frac{1}{x^2} \end{cases}$$

$$\text{Norm:} \begin{cases} f(x) &= \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2\sigma^2} (X-\mu)^2} \\ \text{Mean} &= \mu \text{ (location), SD} = \sigma \text{ (shape)} \end{cases}$$

$$Z_\alpha = 100(1 - \alpha)^{\text{th}} \text{prcnt}$$

$$(X - \mu) \div \sigma = Z \sim N(0, 1)$$

$$\sigma Z + \mu = X \sim N(\mu, \sigma)$$

Emp. Rule: $\bar{x} \pm k \times \text{SD}_{(64:1,95:2,99.7:3)}$

$$\text{Zsco: } z = \frac{x - \bar{x}}{s} (>3 \text{ is out})$$

$$\mu_{\bar{X}} = \mu$$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) (\text{giv Or CLT})$$

B: Trials have exactly 2 outcomes, prob of succ. same $\forall$ trials, n fixed in advnc, trials indepndnt

Poi: for prob a # of occ. hppns in gvn spctme intrvl (X). expected # of occ same, total # of occ not fixed, occ. indepndnt

G: # of independent trials until the first success with prob. of success p

HG: X : # of objs w trait outof n whre: m objs outof N have trait, no rplcmnt

Exp: whn looking at amnt of time till event occurs, λ : mean num of occurences during given intrvl

$E(X)$: aka mean of r.v.; analogue for cont. is pdf

B v. Poi: B has finite number of trials, poi does not!

tip: on cont, esp poi, remember complementing to get finite addition problems

tip: remember symmetry, esp. of the t -table (flip and neg.)